user searches for the keyword *vaccine*, the search engine not only retrieves documents related to vaccination but also throws in relevant documents about inoculations, jabs, and other vaccine-related terms.

Elasticsearch employs a similarity algorithm to generate a relevance score for full-text queries. The score is a positive floating-point number attached to the results, with the highest-scored document indicating more relevance to the query criteria.

Elasticsearch handles both structured and unstructured data efficiently. One of its key features is its ability to index and search both structured and unstructured data in the same index. This allows us to search and analyze both types of data together and gain insights that would be difficult to obtain otherwise.

1.2.2 Search supported by a database

Old-fashioned search was mostly based on traditional relational databases. Older search engines were based on layered architectures implemented in multi-tiered applications, as shown in figure 1.1.

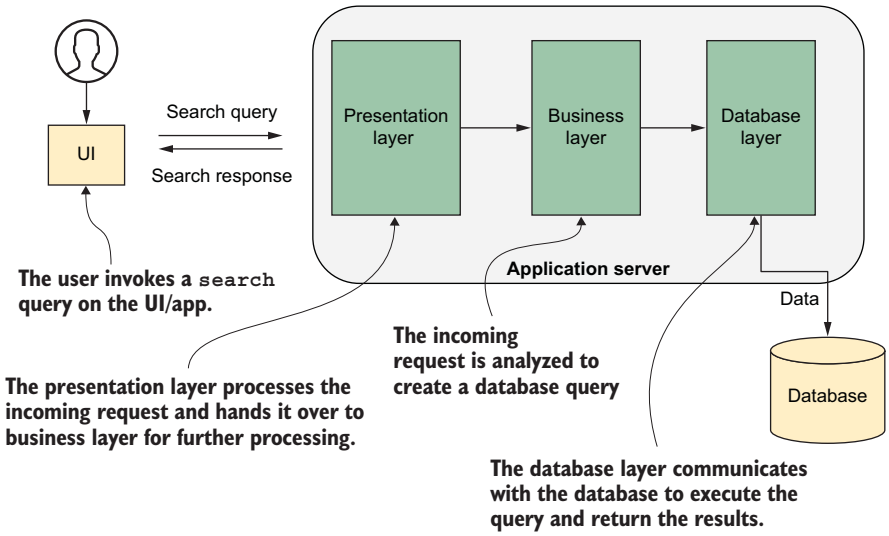


Figure 1.1 Search based on a traditional database

Queries written in SQL using clauses like *where* and *like* provided the foundation for search. These solutions are not necessarily performant and efficient for searching full-text data to provide modern search functionality.

Having said that, some modern databases (Oracle and MySQL, for example) support full-text searching (queries against free text like a blog post, movie review, research article, etc.), but they may struggle to provide efficient searches in near-real time on heavy loads. See the sidebar "Full-text searching with databases" for more details.